

JAVA PRACTICAL 1

1. Count Total Number of Odd Numbers Between 1–500

```
public class OddNumbers {
    public static void main(String[] args) {
        int count = 0;

        for (int i = 1; i <= 500; i++) {
            if (i % 2 != 0) {
                count++;
            }
        }

        System.out.println("Total odd numbers = " + count);
    }
}
```

2. Generate Prime Numbers from 1 to 500

```
public class PrimeNumbers {
    public static void main(String[] args) {
        for (int i = 2; i <= 500; i++) {
            int count = 0;

            for (int j = 1; j <= i; j++) {
                if (i % j == 0) {
                    count++;
                }
            }

            if (count == 2) {
                System.out.print(i + " ");
            }
        }
    }
}
```

3. Find the Sum of Digits of a Number

```
import java.util.Scanner;

public class SumOfDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;

        while (n > 0) {
```

```

        sum = sum + n % 10;
        n = n / 10;
    }

    System.out.println("Sum of digits = " + sum);
}
}

```

4. Check Armstrong Number

```

import java.util.Scanner;

public class Armstrong {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int temp = n;
        int digits = 0;

        while (temp > 0) {
            digits++;
            temp = temp / 10;
        }

        temp = n;
        int sum = 0;

        while (temp > 0) {
            int digit = temp % 10;
            int power = 1;

            for (int i = 1; i <= digits; i++) {
                power = power * digit;
            }

            sum = sum + power;
            temp = temp / 10;
        }

        if (sum == n)
            System.out.println("Armstrong number");
        else
            System.out.println("Not an Armstrong number");
    }
}

```

5. Swap Two Numbers Without Using Third Variable

```

import java.util.Scanner;

public class SwapNumbers {

```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter first number: ");
    int a = sc.nextInt();

    System.out.print("Enter second number: ");
    int b = sc.nextInt();

    a = a + b;
    b = a - b;
    a = a - b;

    System.out.println("After swapping:");
    System.out.println("First number = " + a);
    System.out.println("Second number = " + b);
}
}

```

6. Check Leap Year

```

import java.util.Scanner;

public class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter year: ");
        int year = sc.nextInt();

        if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
            System.out.println("Leap Year");
        else
            System.out.println("Not a Leap Year");
    }
}

```

7. Display Fibonacci Series

```

import java.util.Scanner;

public class Fibonacci {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();

        int a = 0;
        int b = 1;

        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");

```

```

        int c = a + b;
        a = b;
        b = c;
    }
}

```

8. Simple Calculator Using Switch Case

```

import java.util.Scanner;

public class Calculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double a = sc.nextDouble();

        System.out.print("Enter second number: ");
        double b = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        char op = sc.next().charAt(0);

        switch (op) {
            case '+':
                System.out.println("Result = " + (a + b));
                break;

            case '-':
                System.out.println("Result = " + (a - b));
                break;

            case '*':
                System.out.println("Result = " + (a * b));
                break;

            case '/':
                System.out.println("Result = " + (a / b));
                break;

            default:
                System.out.println("Invalid operator");
        }
    }
}

```

9. Count Number of Vowels and Consonants in a Sentence

```

import java.util.Scanner;

public class VowelsConsonants {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter a sentence: ");
        String str = sc.nextLine();

        int vowels = 0;
        int consonants = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = Character.toLowerCase(str.charAt(i));

            if (ch >= 'a' && ch <= 'z') {
                if (ch == 'a' || ch == 'e' || ch == 'i' ||
                    ch == 'o' || ch == 'u')
                    vowels++;
                else
                    consonants++;
            }
        }

        System.out.println("Vowels = " + vowels);
        System.out.println("Consonants = " + consonants);
    }
}

```

10. Check Whether a String is Palindrome

```

import java.util.Scanner;

public class StringPalindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            rev = rev + str.charAt(i);
        }

        if (str.equalsIgnoreCase(rev))
            System.out.println("Palindrome");
        else
            System.out.println("Not a Palindrome");
    }
}

```

11. Check Whether Two Strings are Anagrams

```

import java.util.Scanner;
import java.util.Arrays;

public class Anagram {

```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter first string: ");
    String str1 = sc.nextLine();

    System.out.print("Enter second string: ");
    String str2 = sc.nextLine();

    str1 = str1.replace(" ", "").toLowerCase();
    str2 = str2.replace(" ", "").toLowerCase();

    char[] a = str1.toCharArray();
    char[] b = str2.toCharArray();

    Arrays.sort(a);
    Arrays.sort(b);

    if (Arrays.equals(a, b))
        System.out.println("Anagram");
    else
        System.out.println("Not an Anagram");
}
}

```

12. Check Strong Number Using Command Prompt

```

public class StrongNumber {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);
        int temp = n;
        int sum = 0;

        while (temp > 0) {
            int digit = temp % 10;
            int fact = 1;

            for (int i = 1; i <= digit; i++) {
                fact = fact * i;
            }

            sum = sum + fact;
            temp = temp / 10;
        }

        if (sum == n)
            System.out.println("Strong number");
        else
            System.out.println("Not a Strong number");
    }
}

```

To run:

```
javac StrongNumber.java
```

```
java StrongNumber 145
```

13. Find $S = 1^2/1! + 2^2/2! + \dots + n^2/n!$

```
import java.util.Scanner;

public class Series1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter n: ");
        int n = sc.nextInt();

        double sum = 0;
        double fact = 1;

        for (int i = 1; i <= n; i++) {
            fact = fact * i;
            sum = sum + ((double) (i * i) / fact);
        }

        System.out.println("Sum = " + sum);
    }
}
```

14. Find $S = 1 - 3 + 5 - 7 + 9 \dots + n$

```
import java.util.Scanner;

public class Series2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter n: ");
        int n = sc.nextInt();

        int sum = 0;
        int sign = 1;

        for (int i = 1; i <= n; i += 2) {
            sum = sum + sign * i;
            sign = sign * -1;
        }

        System.out.println("Sum = " + sum);
    }
}
```

15. Convert a Number to Words Using Switch

```
import java.util.Scanner;

public class NumberToWords {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a number: ");
    String n = sc.next();

    for (int i = 0; i < n.length(); i++) {
        char ch = n.charAt(i);

        switch (ch) {
            case '0':
                System.out.print("zero ");
                break;
            case '1':
                System.out.print("one ");
                break;
            case '2':
                System.out.print("two ");
                break;
            case '3':
                System.out.print("three ");
                break;
            case '4':
                System.out.print("four ");
                break;
            case '5':
                System.out.print("five ");
                break;
            case '6':
                System.out.print("six ");
                break;
            case '7':
                System.out.print("seven ");
                break;
            case '8':
                System.out.print("eight ");
                break;
            case '9':
                System.out.print("nine ");
                break;
        }
    }
}

```